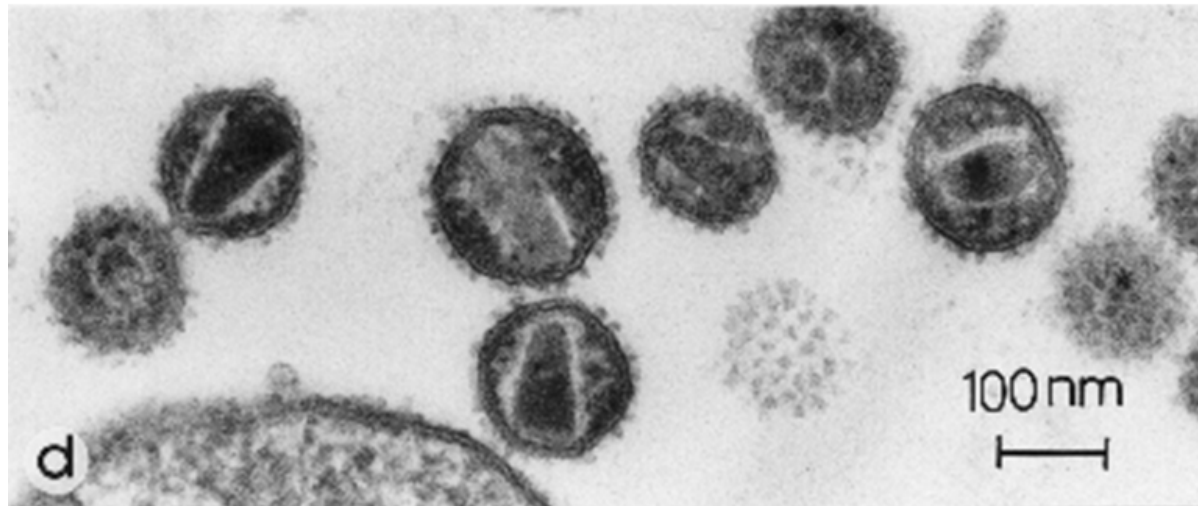


Inferring Genetic Diversity from Next-generation Sequencing Data

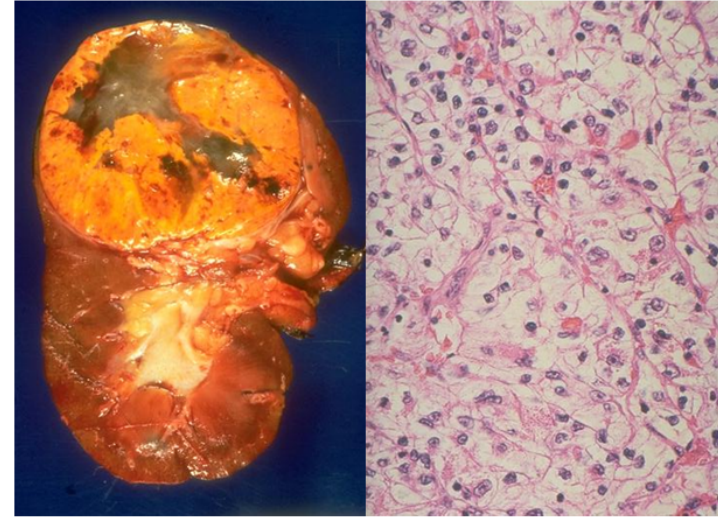
Some slides adapted from an ECCB Tutorial, together with Karin Metzner (UHZ) and Niko Beerenwinkel (ETHZ)



HR Gelderblom *et al.*, Virology 1987

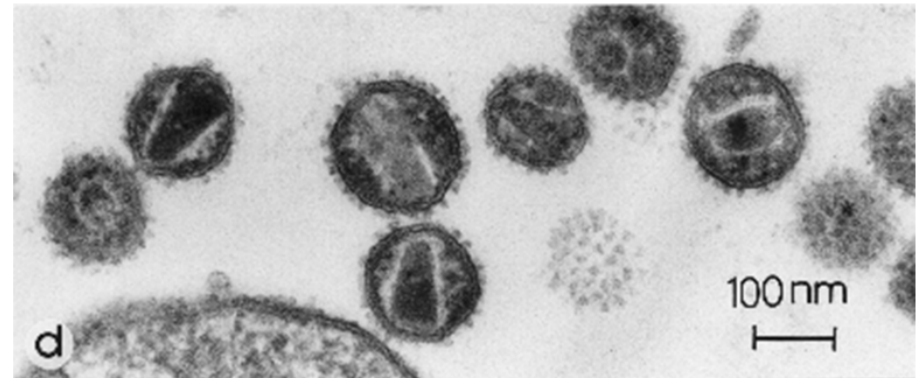
Diversity within an organism

- Genetic diversity in an organism:
intra-tumour diversity



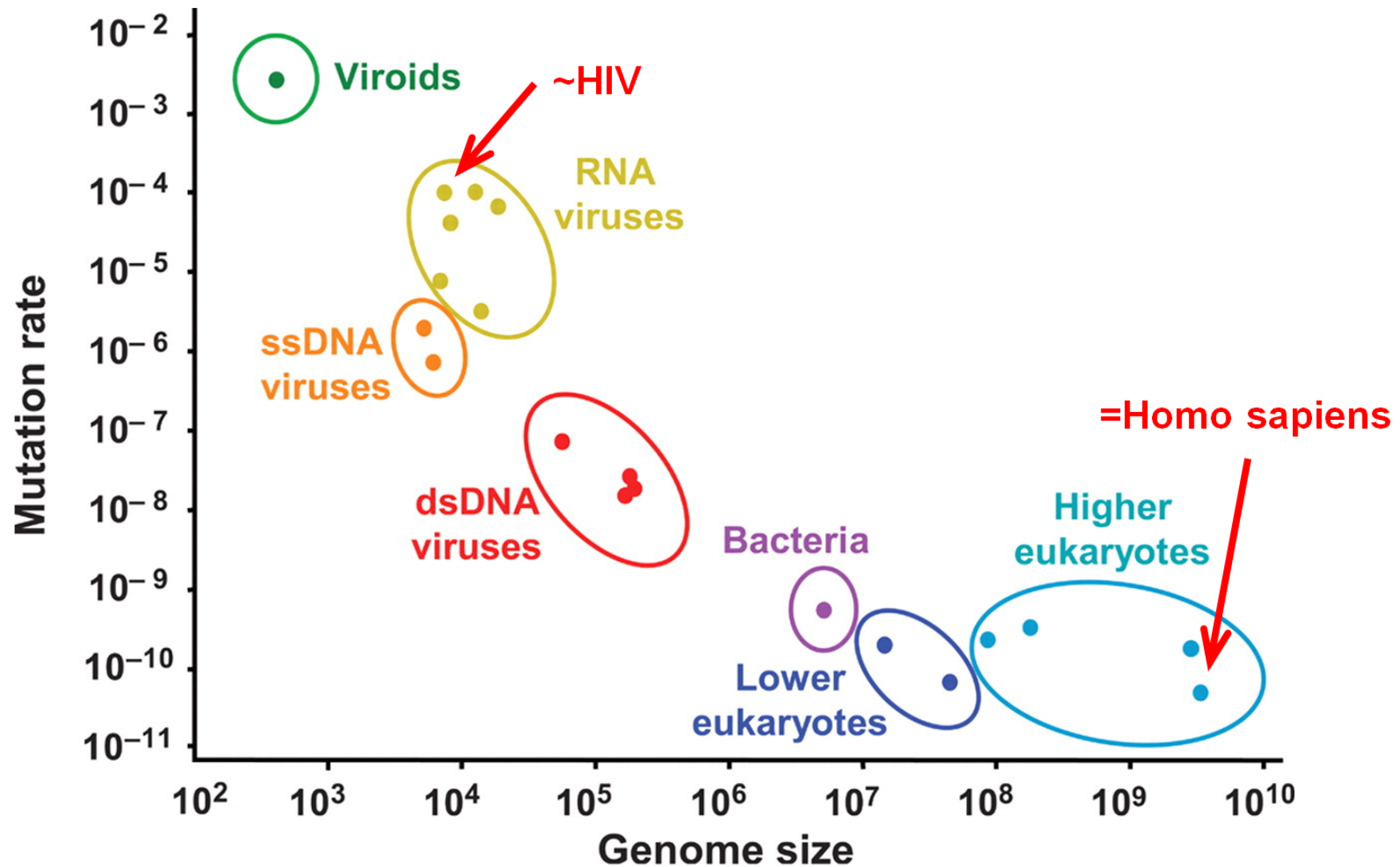
Copyright 2004-2012 University of Washington, UW Medicine Pathology

- Genetic diversity of a species in a defined environment:
intra-patient diversity of HIV-1



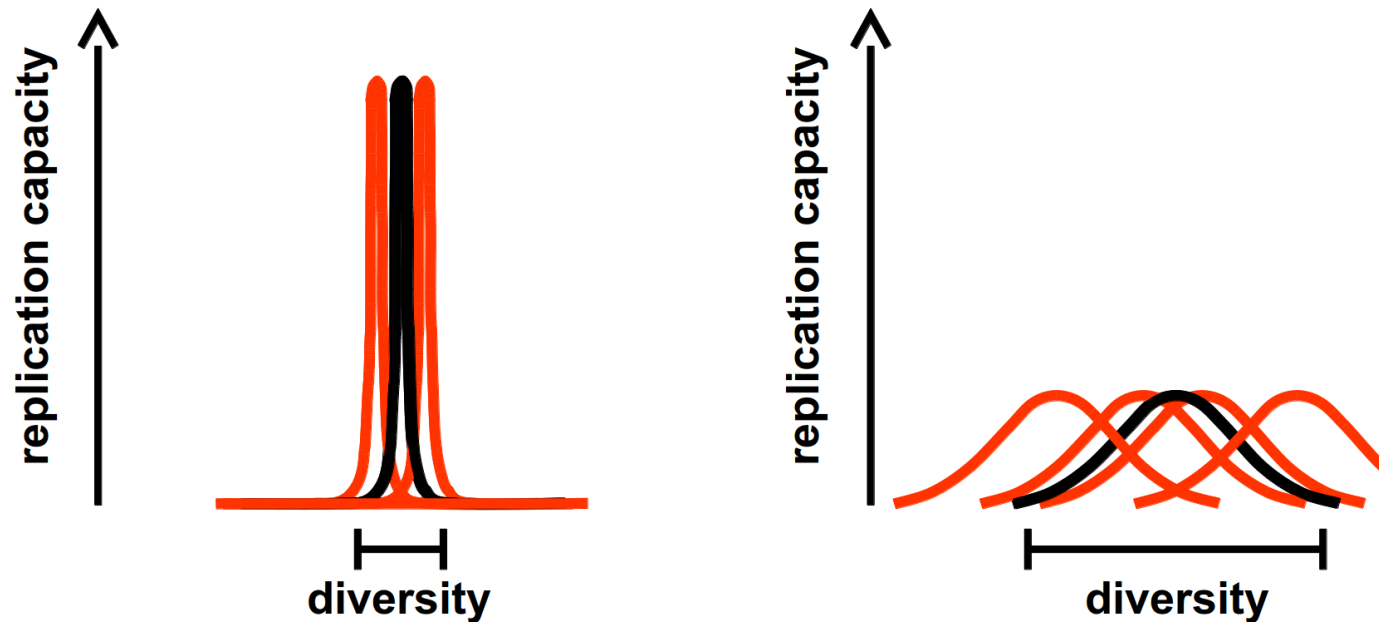
HR Gelderblom *et al.*, Virology 1987

Mutation rate correlates with genome size



adapted from S Gago *et al.*, Science 2009

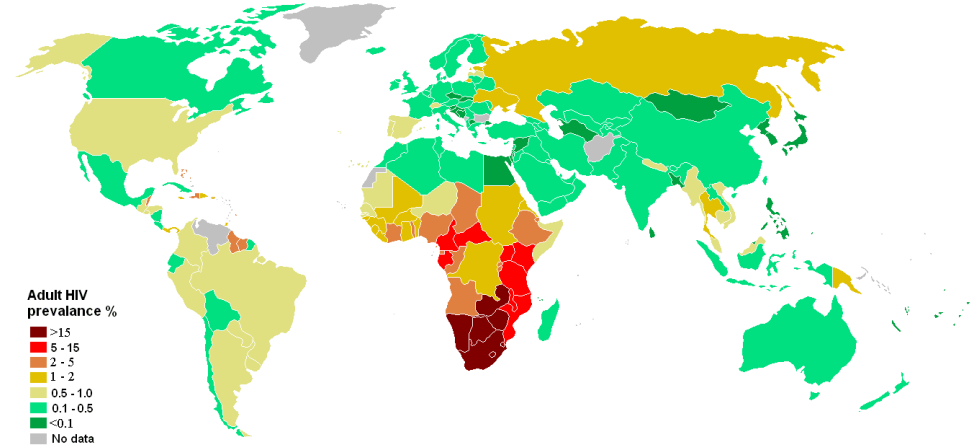
Viral quasispecies



- **Quasispecies model of viral evolution** (Eigen & Schuster, 1979)
- Selection pressure on the whole population rather than on single individual: generation of a **broad quasispecies** with members of **approximately equal fitness** is a promising evolutionary strategy.
- Viral quasispecies = viral population = mutant cloud = swarm
Virus variant = viral haplotype

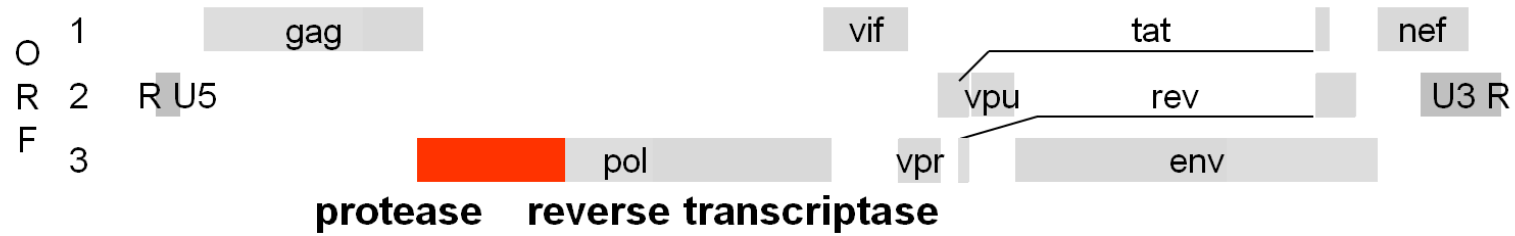
HIV and AIDS

- Acquired Immunodeficiency Syndrome (AIDS) is caused by Human Immunodeficiency Virus (HIV). More than 35 Million people are affected world-wide.



- HIV **quasispecies** are a set of genetically diverse (**haplotypes**).
- HIV has **high mutagenicity**
 - ~ **evolutionary escape** from the host's immune response
 - ~ **drug resistance**.
- Identify this genetic diversity to enable **Personalized Medication**.

HIV drug resistance



RT

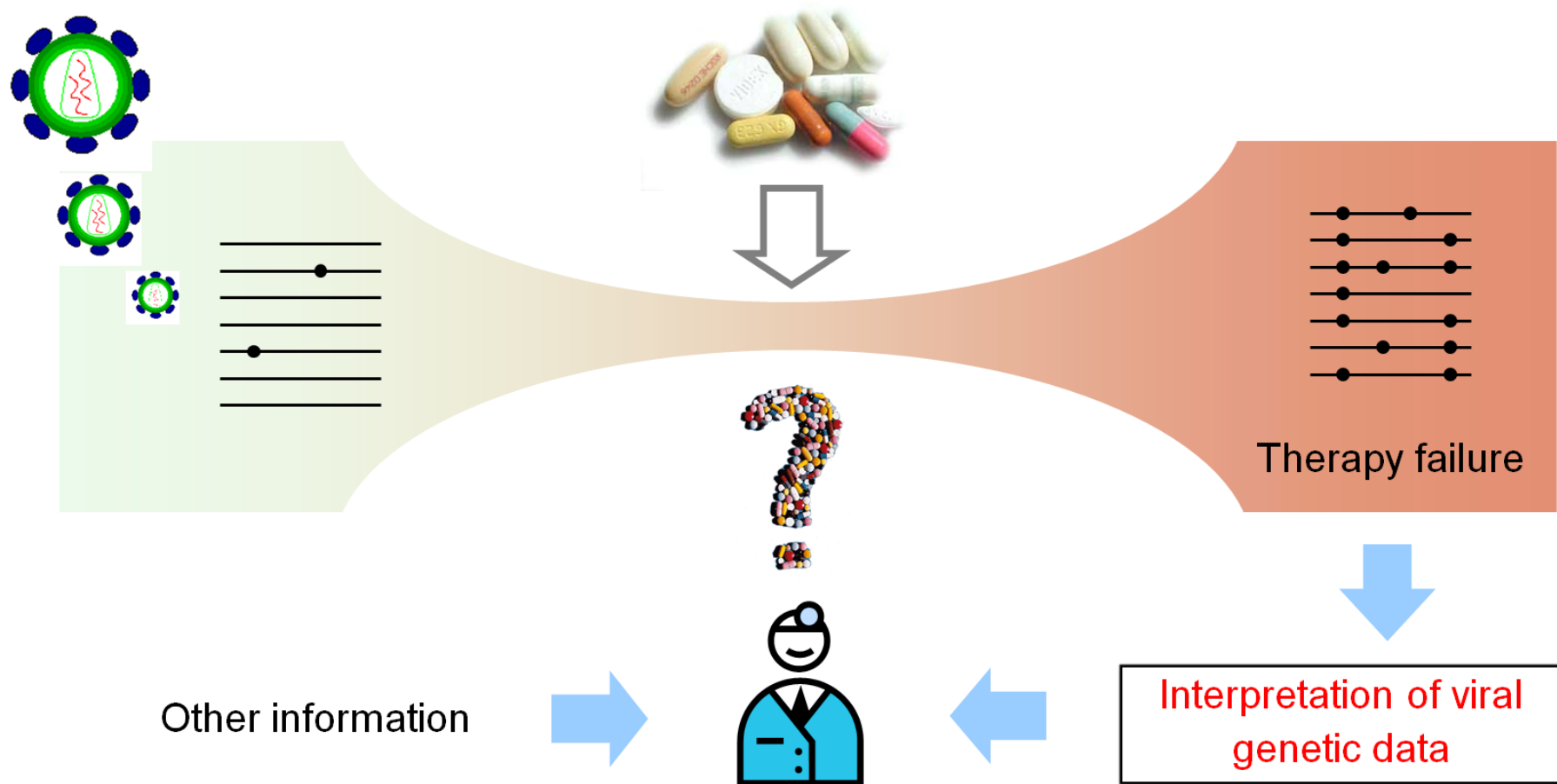
Lamivudine	K 65 R	M 184 V I				
Zidovudine ^{d,e,j,k}	M 41 L	D 67 N	K 70 R	L 210 W	T 215 Y F	K 219 Q E

PR

Lopinavir/ ritonavir ^u	L 10 F I R V	K 20 M R	L 24 I	V 32 I	L 33 F	M 46 I L	I 47 V A	I 50 V	F 53 L	I 54 V L A M T S	L 63 P	A 71 V T	G 73 S	L 76 V	V 82 A F T S	I 84 V	L 90 M
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VA Johnson *et al.*, Top HIV Med 2009

HIV drug resistance, individualized treatment

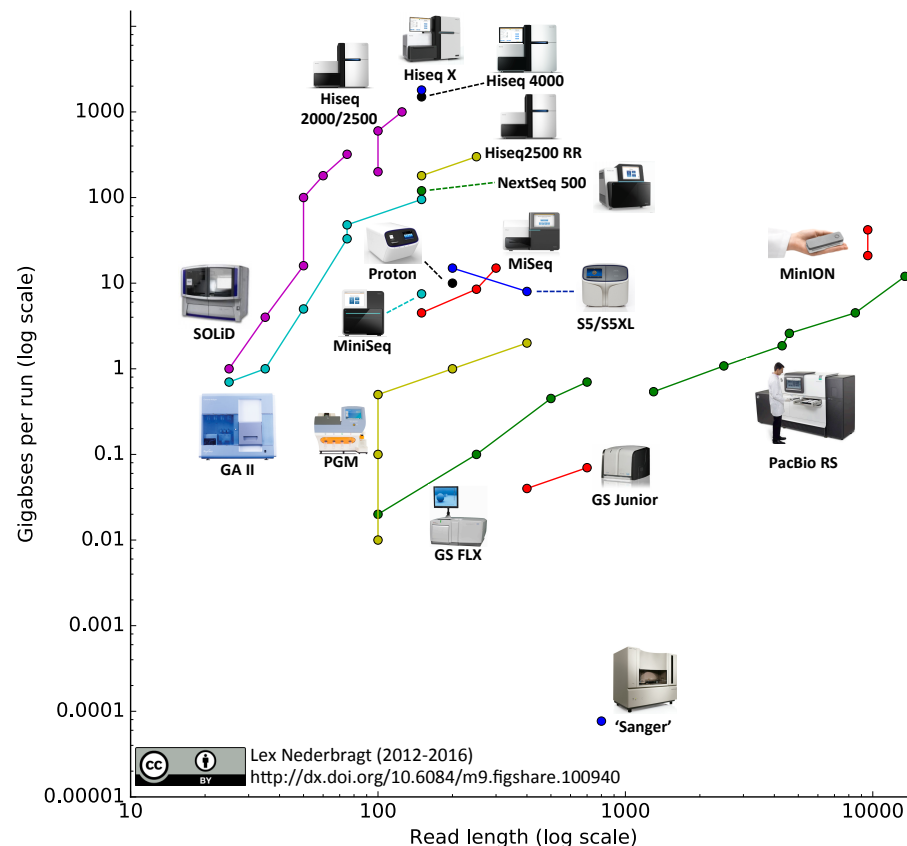


HIV evolution is **highly responsive to selection pressure** from drugs that are not fully suppressive

~> **rapid outgrowth of resistant variants.**

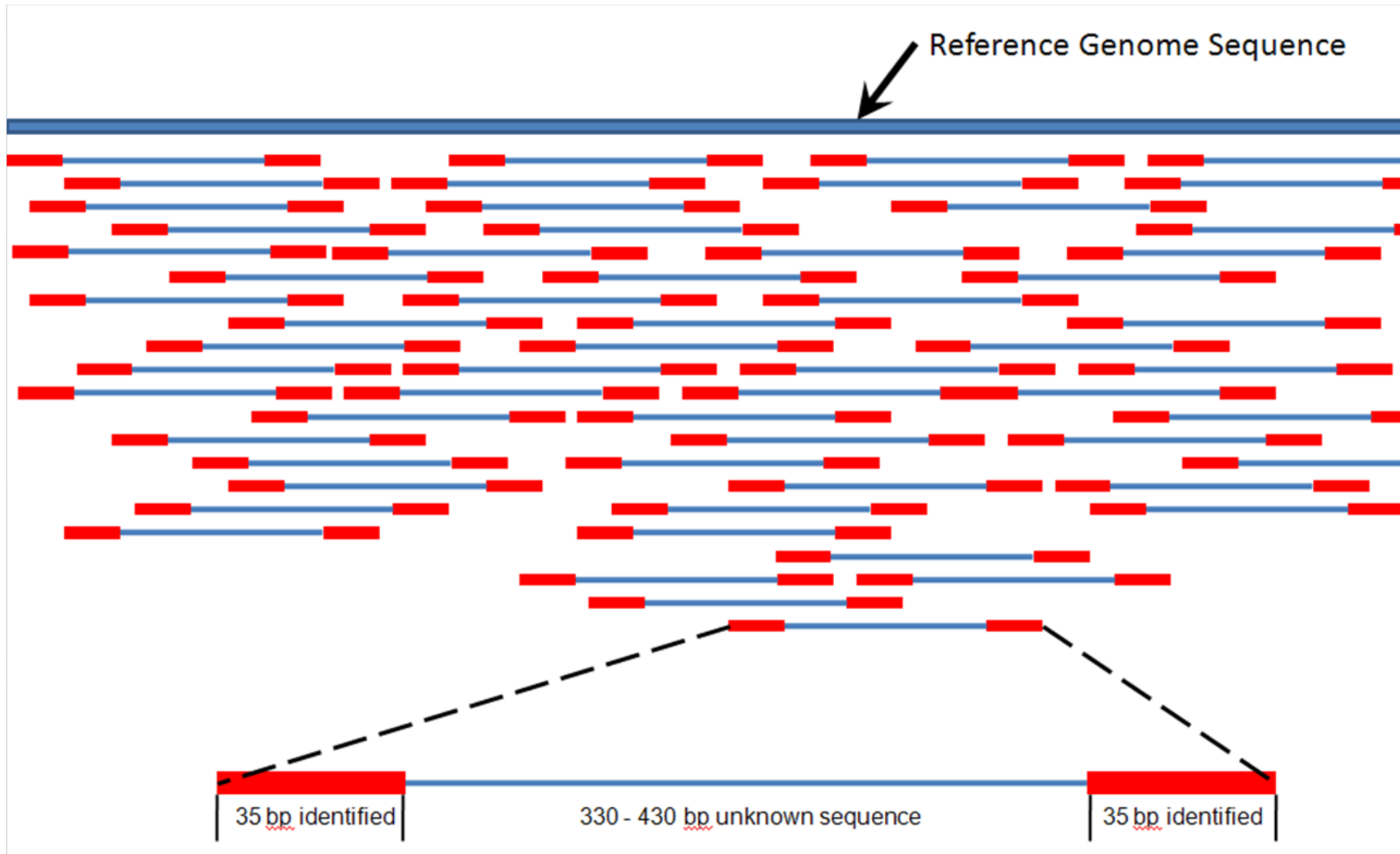
DNA Sequencing: Shotgun sequencing

- **Shotgun sequencing:** Sequencing DNA strands by way of randomized fragmentation (analogy with quasi-random firing pattern of a shotgun).
- **Short-read or next-generation** sequencing produces short sequences (“reads”) (25-500 bp), but many of them ($10^5 - 10^7$ per day).



Nederbragt, Lex (2016): developments in NGS. figshare. Dataset. <https://doi.org/10.6084/m9.figshare.100940.v9>

Mapping short reads to reference genome

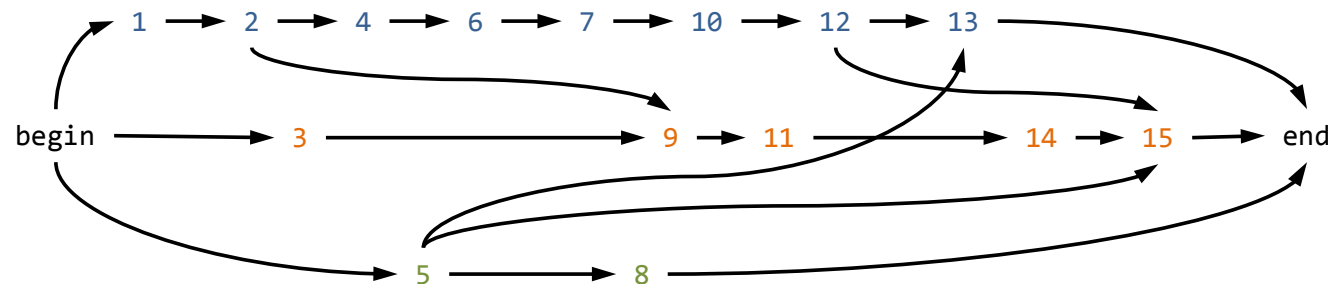


By Suspencewl - Own work, CC0, <https://commons.wikimedia.org/w/index.php?curid=13764860>

Combinatorial Assembly: Read Graph

↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓ ↓
 A CCTGAAATCACTCTATGGCAACGACCCATCGTCACAATAAAGATAGGG 60%
 B CCTCAAATCACTCTTTGGCAACGACGCATCGTCACAATATAGATAGGA 30%
 C CCTCAAATCTCTCTTTGGCACCGACCCATCGTCCCAATAAAGATAGGG 10%

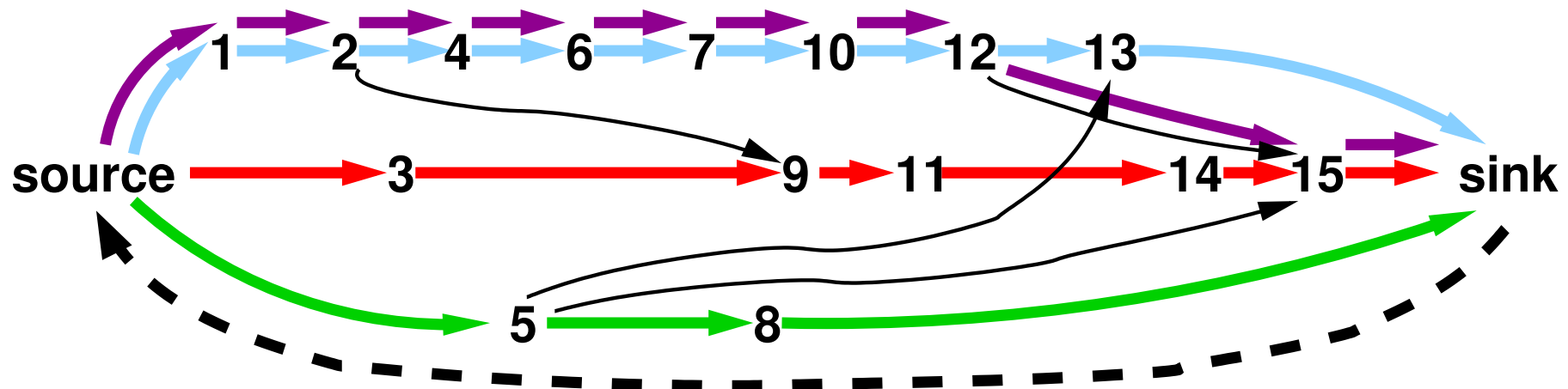
1 CCTGAAATCACTCTATGGCA
 2 GAAATCACTCTATGGCAACG
 3 ATCACTCTTTGGCAACGACG
 4 TCACTCTATGGCAACGACCC
 5 CTCTCTTTGGCACCGACCCA
 6 CTATGGCAACGACCCATCGT
 7 TATGGCAACGACCCATCGTC
 8 TTGGCACCGACCCATCGTCC
 9 TGGCAACGACGCATCGTCAC
 10 CAACGACCCATCGTCACAAT
 11 CAACGACGCATCGTCACAAT
 12 AACGACCCATCGTCACAATA
 13 CGACCCATCGTCACAATAAA
 14 GCATCGTCACAATATAGATA
 15 CATCGTCACAATATAGATAG



Each path is a potential haplotype

Network Flow

- A HT h corresponds to a **path from source to sink** in the read graph.
- Each path can be viewed as a **flow** $source \rightarrow \{\text{reads from } h\} \rightarrow \text{sink}$.
- The value of the (circular) **flow** f through a read is the **number of haplotypes that contain the read**.
- **Main idea:** minimizing flow $\leadsto$ most parsimonious HT assembly.



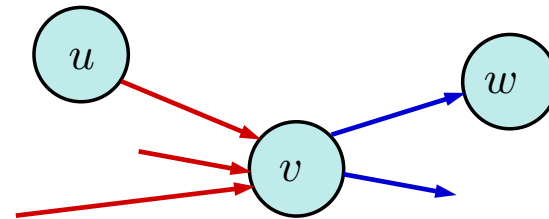
LP for Most Parsimonious Quasispecies Assembly:

Main idea: Every unit of flow from a single HT should pass through (sink, source)-edge only **once**.

Objective: Minimize **total flow** (=backflow) $f(\text{sink}, \text{source})$

Subject to:

- **Flow conservation:**



$$\forall v \in V : \sum_{(u,v) \in E} f(u, v) = \sum_{(v,w) \in E} f(v, w)$$

- **Each read covered by at least one haplotype:**

$$\forall v \in V : \sum_{(u,v) \in E} f(u, v) \geq 1.$$

- **Physical constraints:** $\forall (u, v) \in E : f(u, v) \geq 0.$